Selective multiple carbonylations of di- and trienes – An efficient way to renewable chemicals and polymers

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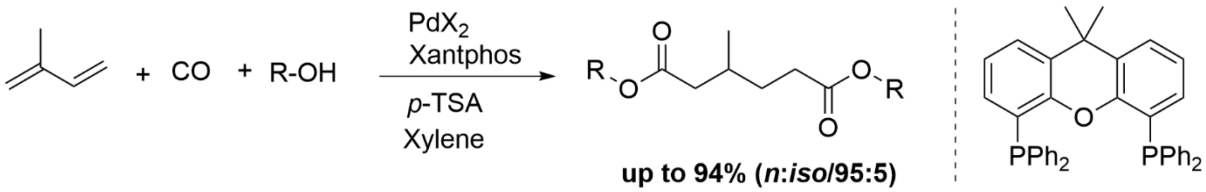
The direct dicarbonylation of 1,3-dienes provides highly atom-economic access to important intermediates for the manufacture of polyamides and polyesters – processes that operate on the multi-million ton-scale. So far, this reaction has been realized only for adipic esters from 1,3-butadiene,[1,2] and no catalytic systems ensuring satisfying activity for other feedstocks including renewables are known. Here, we present for the first time a catalytic system that allows dicarbonylation of isoprene and numerous other dienes in excellent yield and selectivity with important implications also for future development of tricarbonylation of trienes for production of alternative and renewable plasticizers and plastics.

In general, the dicarbonylation of dienes and trienes is hindered by the strong coordination of the substrates to the catalyst, ultimately leading to low or no activities. Careful balancing of the ligand environment by choosing the right ligand and solvent is crucial to enable the desired transformation.



Scheme 1. Dicarbonylation of isoprene enabled by a Pd/Xantphos-based system.

The herein presented catalyst system relies on Xantphos and Xantphos-like ligands in the presence of specific counter anions (Scheme 1). Notably, aromatic solvents significantly enhance the system's activity and are far superior compared to other solvents. Applications of this novel system to carbonylations of (bio-based) trienes allows for the synthesis of a renewable triesters via direct tricarbonylation.

Literature

[1] J. Yang, J. Liu, H. Neumann, R. Franke, R. Jackstell, M. Beller, *Science* **2019**, *366*, 1514-1517.

[2] J. Yang, J. Liu, Y. Ge, W. Huang, F. Ferretti, H. Neumann, H. Jiao, R. Franke, R. Jackstell, M. Beller, *Angew. Chem. Int. Ed.* **2021**, *133*, 9613-9619.